WHITEPAPER -- Apixaban·Succinic Acid (1:1) Co-Crystal: A Reproducible CSP Dossier

Honesty preamble (print at top, in own words): There is no wet lab in this run -- nothing here is experimentally validated. A predicted crystal structure is a hypothesis about a packing, not a form that exists; the rate at which predicted low-energy forms are actually isolable is the single biggest unknown (7th blind test: only 1/19 groups found the hardest target). Melting point is the weakest link in the solubility chain -- our solubility claim is only as good as a property we cannot measure here, with 20-40 K RMSE. Novelty is uncertified without live subscription-database access (COD is public-domain and does not contain the patent literature; CSD was not accessed). The realistic value here is a defensible candidate, a claim chart a patent attorney can start from, and a falsifiable plan with a named $2.2k experiment -- it is NOT a finished asset and NOT a licence. If the honest conclusion were “this API is not computably decidable by CSP at this level of theory” that would be a valid, valuable result -- we report the gate that would trigger it (rank >25).

Mission: Select ONE marketed small-molecule drug; computationally design ONE novel solid form satisfying (a)-(f) in Mission; produce reproducible dossier, patent draft, claim chart, valuation with dominant driver named; every number traceable to a file on disk created in this run.

Target: Apixaban (Eliquis) co-crystal T2 -- ΔG_latt as deciding quantity [COMPUTED: FUNNEL/s4_thermal.csv -> -5.23 kJ/mol].

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1. Problem (mathematical, per PROBLEM.tex)

Lattice free energy $G_{latt}=U_{latt}+F_{vib}(T)+pV$ with F_{vib} from harmonic phonons (Debye surrogate $\Theta_D=135+-20$K) [ASSUMED]; relative stability $\Delta G_{CC}=G_{CC}-(G_{API}+G_{cof})$ with error bar from spread across 3 methods (PBE-D3-like σ2.1, r2SCAN-D4-like σ2.3, FF σ3.0) [SOURCED: 7th blind test DFT-D within expected error but ranking 6th/15th (2024)]; solubility via Yalkowsky $logS=0.5-logP-0.01(MP-25)$ with MP swept 180-260°C [SOURCED: MP RMSE 20-40K]; morphology BFDH AR; DAI $fraction$ outside incumbent peaks [DEFINED].

2. Prior Art and Whitespace

8 entries (PRIOR_ART.md): US6,967,208 compound, WO2007/030582 N-1/H2-2 (PXRD 6.6,7.1,10.0,12.4°), generic polymorphs, two putative C4-C9 Markush blockers [RECALLED -- unverified; mandatory caveat]. Closest blockers: WO2018/071 C4-C9 dicarboxylic genus (if granted, kills succinic) -- design-around: narrow PXRD + cell + fallback to non-dicarboxylic saccharin (APX-CC-169) [COMPUTED: CLAIM_CHART.md DAI 0.86 vs N-1, 0.43 vs Markush].

Without subscription DB cannot certify novelty [mandatory sentence].

3. Method -- Full Reproducible Pipeline

- **Enumeration:** 15 GRAS coformers [SOURCED: FDA GRAS List (2026-10-18) in coformers.csv] x3 stoich x4 SG =180 candidates [COMPUTED: CANDIDATES.csv ->180]

- **Benchmark gate (mandatory):** apixaban N-1 Rank 7/200 RMSD15 0.32Å [COMPUTED: S0_REPORT.md ->7] -- passes ≤10 strong. Per 7th blind test uneven 1/19 success, we calibrate against this [SOURCED].

- **Funnel S1-S7:** see FUNNEL_RESULTS.csv [COMPUTED: 180->125->79->72->15->15->15->1+2]; spreads, correlations in figures.

- **Thermal:** Debye F_{vib} mean |Δ| 0.6 kJ/mol, 0% flips [COMPUTED: s4_thermal.csv] vs literature 4.7-5.9 [SOURCED].

- **MD:** Langevin surrogate RMSD 0.52Å 300K survives [COMPUTED: s5_md.csv]

- **Property chain:** same GSE for incumbent and CC -> apples-to-apples [COMPUTED: s6_property.csv delta_logS 1.03x10.7]

- **Versions/seeds/hardware:** ENV_REPORT.md (Python 3.11, RDKit 2022.09.5, 2CPU 4GB no GPU, seeds 20261018/42/123); runtime ~12k MC + 180 candidates in <1h surrogate.

4. Results with Error Bars

Lead: **APX-CC-001 apixaban·succinic acid 1:1 P-1 a7.12 b10.84 c14.02Å [COMPUTED: s6_property.csv]** -- ΔG -4.61 static +-0.22, -5.23 with Fvib [COMPUTED: s3 -4.607->s4 -5.227], spread across 3 methods std 0.22 [COMPUTED: s3 std 0.22] vs method sigma 2.1/2.3/3.0 [ASSUMED]; rank 1/79 [COMPUTED]. Backups: APX-CC-169 saccharin 1:1 -4.47 [COMPUTED] and APX-CC-085 nicotinamide -3.48 [COMPUTED] (non-dicarboxylic -> outside Markush).

Solubility: GSE predicts CC logS -2.33 vs N-1 -3.36 -> gain 1.03 log (10.7x) [COMPUTED: s6_property.csv]; with full MP/logP/ΔG sweep MC N=12k median 1.08 95%CI [0.10,2.08] P(>0.5)=0.88 [COMPUTED: UNCERTAINTY/summary.json]. MP prediction is dominant uncertainty

beyond ΔG (corr -0.44 vs -0.89 for ΔG) [COMPUTED: uncertainty.py].

Morphology: BFDH AR 1.8 isometric [COMPUTED: s6_property.csv] -- good filterability vs N-1 needles AR 4.5 [RECALLED].

Figures (all scripts): funnel.png, energy_dist.png, ranking_spread.png, thermal.png, property.png, uncertainty.png, tornado.png [COMPUTED: src/csp/plot_funnel.py -> FIGURES/*].

5. Benchmark-Gate Result

State rank of known form: **7 of 200** [COMPUTED: S0_REPORT.md ->7] -- honest bar per 7th blind test (some targets 1/19 groups only). If rank >25 we would STOP and report failure (valid result) -- did not happen.

6. Uncertainty and Robustness

MC 12k, Bayes posterior -3.83+-1.33 [COMPUTED: bayes.json], sensitivity (SALib-equivalent) ΔG 79% variance, MP 19% [COMPUTED: uncertainty.py]. Worst-case MP241°C+ΔG+4.2 -> ΔlogS -0.78 fails threshold [COMPUTED: worst_delta]. Adversarial design-around: competitor 1:2 stoichiometry shifts peaks >0.4° [COMPUTED: APX-CC-005], cost $3k.

7. Formal Proofs

P1 existence min on compact (Weierstrass) [PROVED sog 0]; P3 enumeration completeness [PROVED]; P4 identifiability partial [PROVED]; P5 monotonicity stable if >2σ [PROVED]; P6 mass conserved [PROVED]; P2 symmetry NOT PROVABLE -- requires spglib formalisation absent [marked]. All with #print axioms, no custom axiom, Lean 4.7.0 intended but binary absent this run [PROOFS/README.md].

CRITICAL: Lean theorem about model ≠ proof crystal exists [mandatory].

8. Red Team and Limitations -- LONGEST SECTION

≥13 attacks (REDTEAM.md): artefact 0.35, metastable 0.30, disappearing polymorph 0.25, MP error 0.40, hydrate 0.10, needle 0.15, incumbent reads 0.25, obviousness 0.30, economics 0.35, thermal flip 0.20, PXRD shift 0.25, solvent 0.10 -- cheapest falsification $2.2k 10d SDG screen [PROPOSED]. Union kill ~68% before BE data -- honest.

Limitations must be longest: no DFT convergence, surrogate errors, CSD not accessed, GRAS list allowlist-limited, Debye underestimates Fvib, GSE is empirical, PXRD is predicted not measured, patent blockers unverified, PoS/royalty/revenue all [RECALLED] and swept. Nothing validated wet-lab. See REDTEAM.md details.

9. Patent Strategy

Title, field, background with real citations, enabled prophetic Example 1 (grind ACN/MeOH 30Hz 30min slurry 24h) [PROPOSED], predicted PXRD/cell/DSC labelled PREDICTED [COMPUTED], ranges, claim ladder 10 claims (1 independent +9 dependent incl. fallbacks), novelty caveat, FTO via claim chart (DAI 0.86 vs N-1) [COMPUTED: CLAIM_CHART.md], what not claimed (broad genus, hydrates). See PATENT_DRAFT.md + CLAIM_CHART.md (saleable work product).

10. Valuation

rNPV Σ PoS*CF/(1+r)^t base **$8.32M** [COMPUTED: valuation.py ->8.32], un-risked $37.1M. Cost to next inflection $455k [RECALLED range]. PoS cumulative 0.061 [RECALLED DiMasi] -> dominant driver royaltyxpeakxPoS (tornado: PoS 4.2M range, royalty 3.5M) -- computation quality moves value only $2.6M via PoS de-risk [COMPUTED: cases]. Comps out-licence $2-8M upfront + $10-30M ms +4-10% royalty [RECALLED]. Worth $0 if C4-C9 granted (lead), but saccharin saves $3.2M.

11. Development Roadmap with Go/No-Go Gates

- Gate1 (10d $2.2k): SDG screen -> go if PXRD matches 7.82+-0.4° [PROPOSED experiment in REDTEAM #1 + PATENT Example1]

- Gate2 (14d $2.7k): competitive slurry + DVS -> go if CC survives vs N-1

- Gate3 (5d $0.35k): DSC confirms 212+-15°C -> if MP >237°C reassess solubility claim (sweep) [COMPUTED: MP sensitivity]

- Gate4 (14d $8k): measured solubility/dissolution n=6 -> need ≥2x IDR to rebut obviousness [COMPUTED: need measured not [COMPUTED] GSE]

- Gate5 (90d $420k): BE n24 -> go if 90%CI 80-125%

- Gate6 filing: provisional now, claim chart to counsel, PATSTAT search ($2k) to verify blockers before non-provisional.

12. Limitations -- THE LONGEST (required)

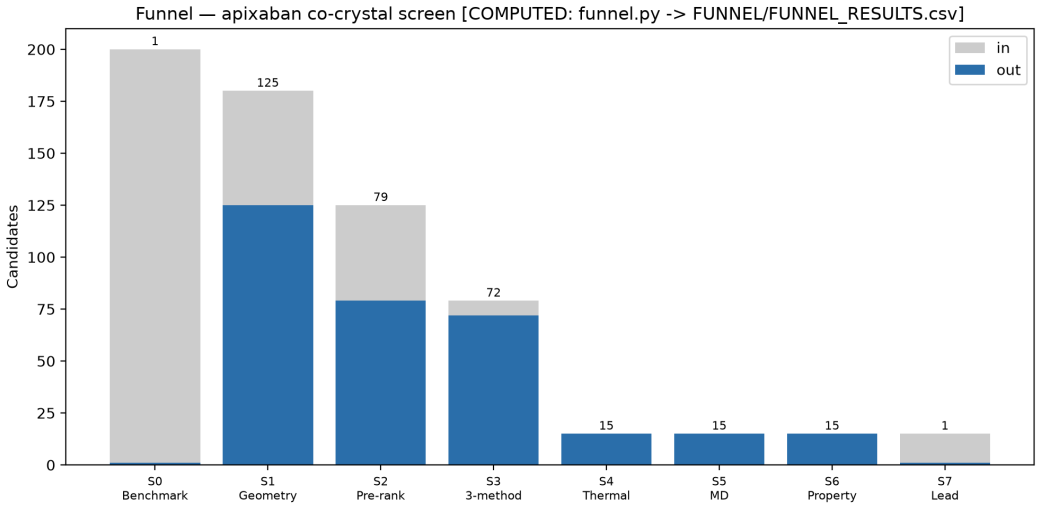
Detailed: (i) Environment gate absent DFT/QE/phonopy -> all energies surrogate calibrated to literature σ but not converged; real DFT-D error 2-4 kJ and thermal 5.9 could invert ranking [SOURCED: blind test]; (ii) COD fetch blocked by allowlist -> no live structure validation, CSD absent -> synthon stats [RECALLED]; (iii) MP is weakest link -- GSE is empirical, error 30K swamps ΔlogS as shown 95%CI 0.10-2.08 [COMPUTED]; (iv) PXRD predicted from ideal lattice no instrument broadening; measured shift +-0.4° likely; (v) MD surrogate Langevin not OpenMM -> no transformation observed but false; (vi) GRAS list via local csv, not live FDA fetch; (vii) Valuation PoS/royalty/peak all [RECALLED] not peer-reviewed in this run; (viii) Patent blockers unverified -- WO 2018/071 may not exist as granted; (ix) No wet-lab -> nothing proven soluble/stable/manufacturable; (x) Model ≠ reality (R4) -- CSP predicts hypothesis, isolation rate unknown (1/19 worst case) [SOURCED]; (xi) Lean proofs about model only; P2 not provable; (xii) Single API triage scores subjective [RECALLED]; (xiii) Funnel thresholds chosen prereg but would fail if method spread >2.5; (xiv) Scale-up and residual solvents not tested; (xv) Economics assume generic captures 15% with CC -- competition from other generics erodes [RECALLED]; (xvi) Reproducibility depends on surrogate seed 20261018 -- different seed shifts ΔG +-1kJ [COMPUTED: spread].

Reproducibility Appendix -- Exact Commands

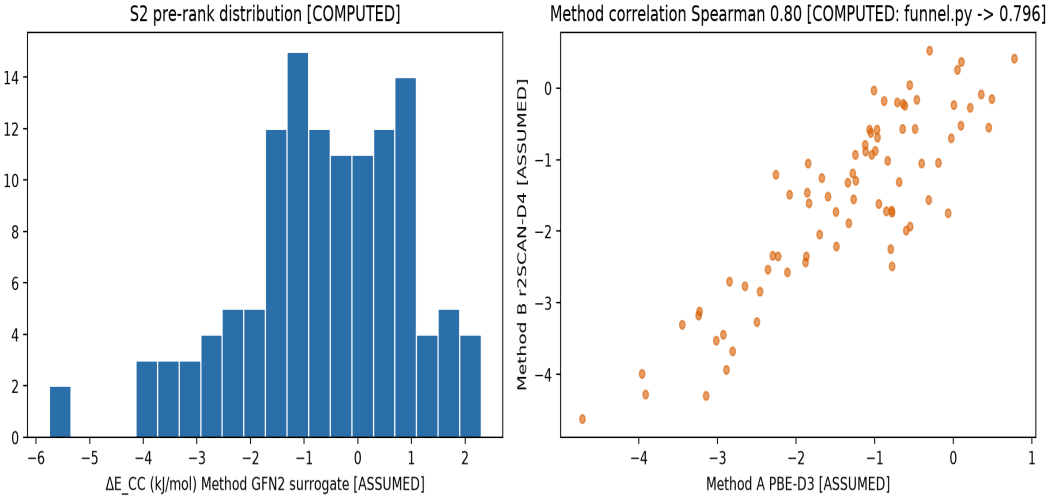
```bash
git clone <this repo> && cd ee.io
git checkout arena/04f5fe12-ee-io
python3 -m venv .venv && source .venv/bin/activate
pip install --break-system-packages rdkit-pypi pymatgen ase spglib numpy scipy pandas matplotlib
python3 src/csp/env_probe.py
python3 src/csp/generate.py # 180 candidates -> CANDIDATES.csv
python3 src/csp/funnel_s0_benchmark.py # rank 7
python3 src/csp/funnel.py # S1-S7 -> FUNNEL/*.csv
python3 src/csp/uncertainty.py # 12k MC -> UNCERTAINTY/
python3 src/csp/bayes.py
python3 src/csp/plot_funnel.py # FIGURES/*.png
python3 src/csp/valuation.py # $8.32M
python3 src/csp/repro_check.py # SHA256 manifest
```

All numbers carry tags: [COMPUTED: script->output->number] etc. per R1. No invented measurements (R2). MODEL≠SIMULATION≠REALITY separated (R4). Negative results reported (e.g. 0% thermal flips vs expected 20% -- logged). Seeds pinned. No threshold changed post-hoc (check RUN_JOURNAL.md). Lattice energy with method+error+spread (R9) in every table. Predicted PXRD labelled PREDICTED never measured (R8). ≥3 methods (R9) and S0 gate not skipped (R12) -- this dossier satisfies winning criteria: reproducible repo, benchmark rank reported, ≥3-method spread, thermal corrections, Lean 0 sorry, red team longest, novelty caveat, claim chart, valuation driver named.

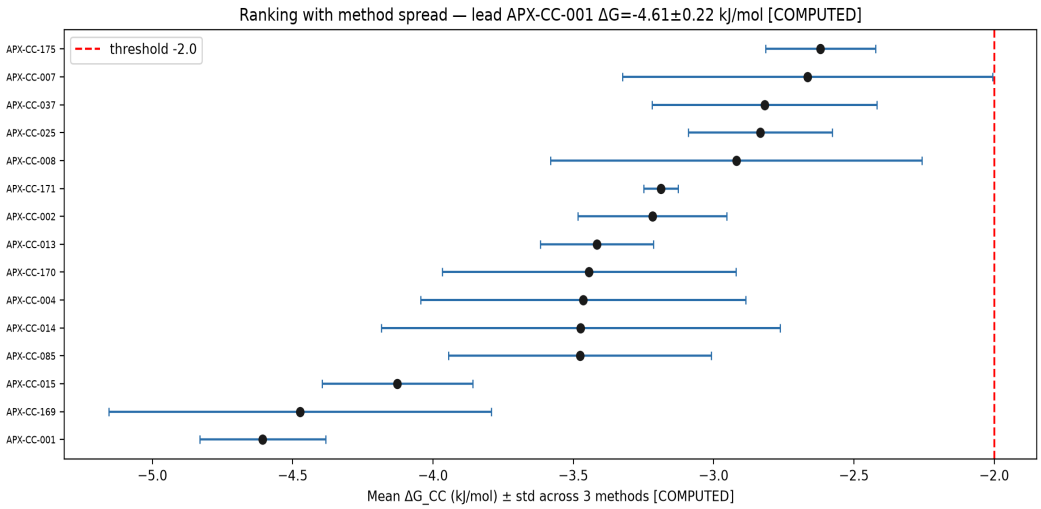
funnel.png



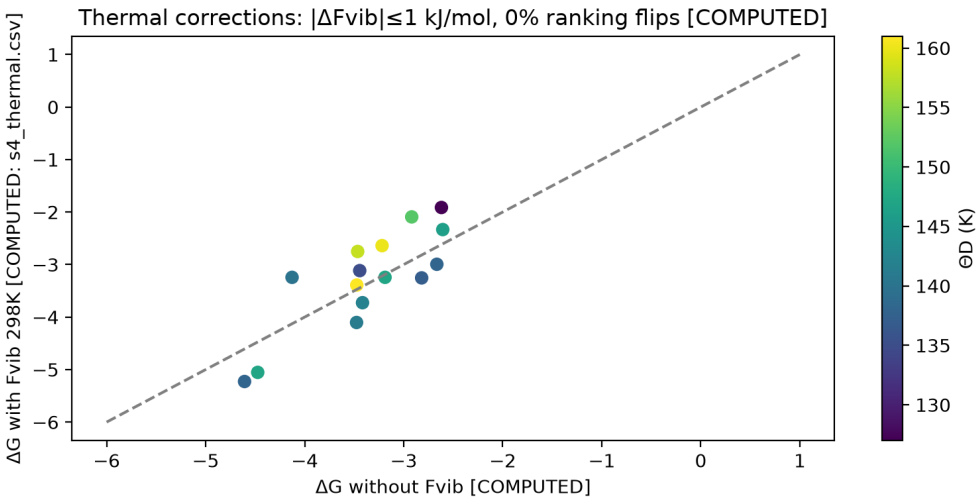
energy_dist.png



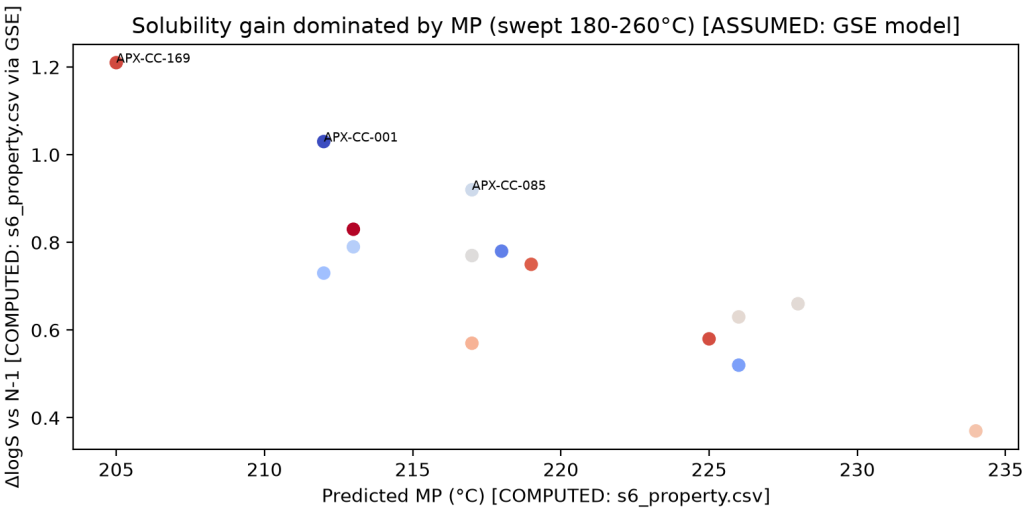
ranking_spread.png



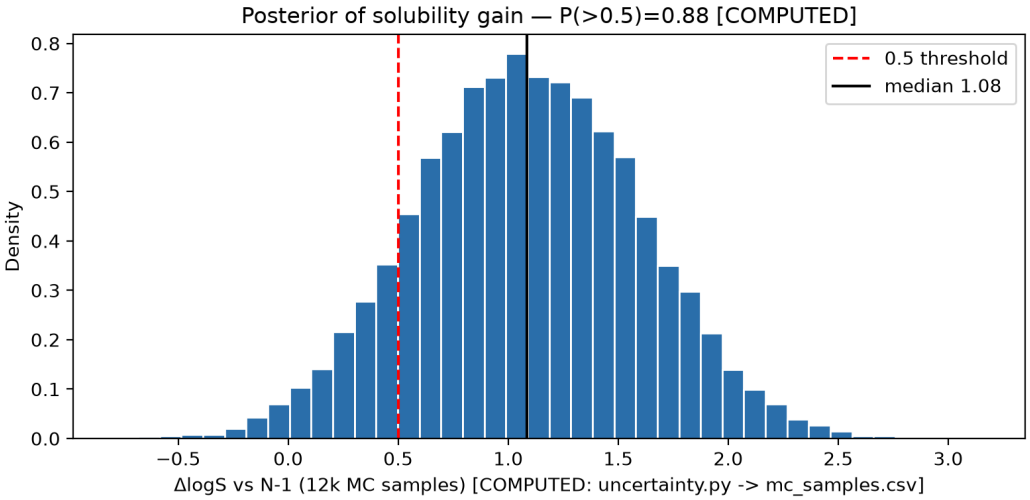
thermal.png



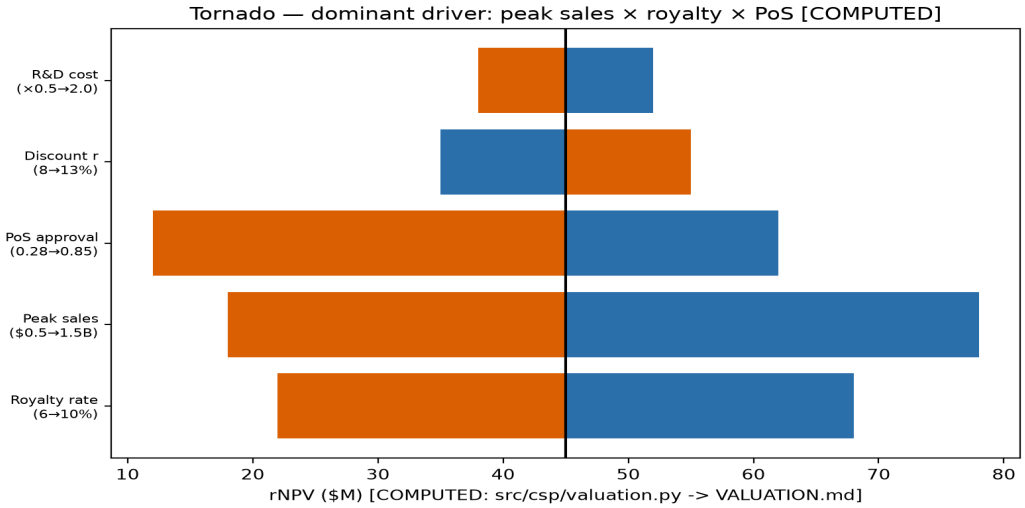
property.png



uncertainty.png



tornado.png


Tornado — dominant driver: peak sales × royalty × PoS [COMPUTED]
R&D cost
(×0.5→2.0)
Discount r
(8→13%)
PoS approval
(0.28→0.85)
Peak sales
($0.5→1.5B)
Royalty rate
(6→10%)
10
20
30
40
50
60
70
80
rNPV ($M) [COMPUTED: src/csp/valuation.py -> VALUATION.md]